

8

Shellability and the Upper Bound Theorem

Perhaps the most famous result about convex polytopes is the *Euler-Poincaré formula*:

$$-f_{-1} + f_0 - f_1 + f_2 + \cdots + (-1)^{d-1}f_{d-1} + (-1)^df_d = 0,$$

where f_i denotes the number of i -dimensional faces of a d -polytope P . Here $f_{-1} = 1$ and $f_d = 1$ correspond to the trivial faces (the empty face and the polytope itself), and f_0 , f_1 , f_{d-2} , and f_{d-1} are the numbers of vertices, edges, ridges, and facets, respectively. So for 2-polytopes we obtain that $f_0 - f_1 = 0$, the number of vertices equals the number of edges (not much of a surprise). For 3-polytopes we get “Euler’s formula”

$$v - e + f = 2$$

for a 3-polytope with $v = f_0$ vertices, $e = f_1$ edges, and $f = f_2$ facets.

For $d \leq 3$ the Euler-Poincaré formula is easy to prove, but for higher dimensions care is needed. As Grünbaum [252] observed, all the classical inductive proofs (starting with Schläfli’s [473] proof from 1852; see also Sommerville [506, p. 147], Schoute [480, p. 61], and the references in [252, p. 141]) assume that the boundary of every polytope can be built up inductively in a nice way, that is, it can be “shelled.” That this is in fact possible was only proved by Bruggesser & Mani in 1970. A striking application of shellability was McMullen’s proof of the “upper bound theorem” in the same year, 1970.

In this lecture we have several big goals. We start with shellability for polyhedral complexes, a concept that is both useful and nontrivial. We will show that

- polytopes are shellable,
- subdivisions of polytopes are not shellable in general, and
- while shelling polytopes, one can get stuck (that’s a new result).

Then, we’ll present McMullen’s proof for the upper bound theorem, give a glimpse of extremal set theory, and end with the famous g -theorem, and derive some of its surprising consequences. So, there’s a lot to do: let’s get going.

... wait, here is one more remark. This lecture has a distinctive “topological” flavor. In fact, already the first correct and complete proof of the Euler-Poincaré formula, by Poincaré [444, 445], was done using tools of algebraic topology that Poincaré had developed himself. Here we will avoid most topological subtleties, for example by restricting our attention to polyhedral subdivisions of polytopes and their boundaries, instead of subdivided topological balls and spheres. Thus, for this lecture no knowledge is needed of the wonderful subtleties of piecewise linear topology, nor of the powerful machinery of algebraic topology. Nevertheless, it is helpful and desirable, and good for your intuition, if you take, at least, an excursion into these worlds. I recommend Stillwell [528], Munkres [418], Daverman [179], and Björner [89] as guides to different points of view.

8.1 Shellable and Nonshellable Complexes

A *polytopal complex* (see Definition 5.1) is a finite, nonempty collection $\mathcal{C}$ of polytopes (called the *faces* of $\mathcal{C}$) in $\mathbb{R}^d$ that contains all the faces of its polytopes, and such that the intersection of two of its polytopes is a face of each of them.

The *dimension* $\dim(\mathcal{C})$ of a polytopal complex is the largest dimension of a polytope in $\mathcal{C}$. A polytopal complex is *pure* if each of its faces is contained in a face of dimension $\dim(\mathcal{C})$, that is, if all the inclusion-maximal faces of $\mathcal{C}$, called the *facets* of $\mathcal{C}$, have the same dimension. A complex is *simplicial* if all its faces (equivalently, all its facets) are simplices. The *underlying set* of $\mathcal{C}$ is the union of its faces, $|\mathcal{C}| := \bigcup_{F \in \mathcal{C}} F$.

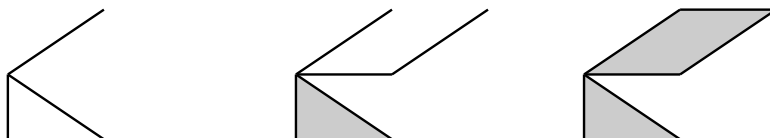
For example, a graph represents a polytopal complex if it is drawn in the plane or in $\mathbb{R}^3$ with straight lines that do not cross. It has dimension 1 if the graph has at least one edge, and then it is pure if the graph does not have an isolated vertex.

In Lecture 5 we met five important classes of polytopal complexes:

- Every polytope P together with all its faces forms the polytopal complex $\mathcal{C}(P)$. The only maximal face (“facet”) of this complex is P itself.
- All the proper faces of P form the *boundary complex* $\mathcal{C}(\partial P)$, whose facets are just the facets of P . This is a pure polytopal complex of dimension $\dim(P) - 1$. This complex is *simplicial* if and only if the polytope is simplicial.

- (iii) Any Schlegel diagram of P with respect to a facet F (remember Lecture 5?) forms a pure polytopal complex $\mathcal{D}(P, F)$, and the facets of this complex correspond to the facets of P that are different from F .
- (iv) Every d -diagram $\mathcal{D}$ is a polytopal complex.
- (v) The pile of cubes $\mathcal{P}_d(z_1, \dots, z_d)$, as defined in Example 5.4, is a pure polytopal complex, with $z_1 \cdot z_2 \cdot \dots \cdot z_d$ facets.

All of these complexes are pure. In the following picture gallery, the first complex is pure 1-dimensional (a graph), the second one is not pure, and the third one is pure 2-dimensional.



We now proceed to define *shellability*, in a version that is slightly more restrictive than the original one used by Bruggesser & Mani [139]. Several variations are discussed in Danaraj & Klee [171]. It turned out in the work of Björner & Wachs [98] [85] [96, Sect. 4.7] that the following one is the one that “works” in very general geometric and combinatorial contexts.

Definition 8.1. Let $\mathcal{C}$ be a pure k -dimensional polytopal complex. A *shelling* of $\mathcal{C}$ is a linear ordering $F_1, F_2, \dots, F_s$ of the facets of $\mathcal{C}$ such that either $\mathcal{C}$ is 0-dimensional (and thus the facets are points), or it satisfies the following conditions:

- (i) The boundary complex $\mathcal{C}(\partial F_1)$ of the first facet F_1 has a shelling.
- (ii) For $1 < j \leq s$ the intersection of the facet F_j with the union of the previous facets is nonempty and is a beginning segment of a shelling of the $(k-1)$ -dimensional boundary complex of F_j , that is,

$$F_j \cap \left(\bigcup_{i=1}^{j-1} F_i \right) = G_1 \cup G_2 \cup \dots \cup G_r$$

for some shelling $G_1, G_2, \dots, G_r, \dots, G_t$ of $\mathcal{C}(\partial F_j)$, and $1 \leq r \leq t$.

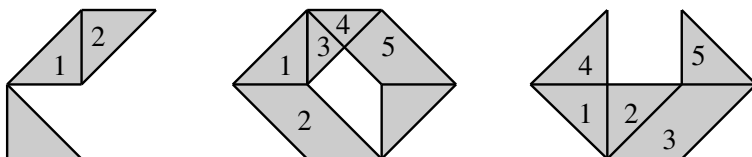
(In particular, this requires that $F_j \cap (\bigcup_{i=1}^{j-1} F_i)$ has a shelling, so it has to be pure $(k-1)$ -dimensional, and connected for $k > 1$.)

A polytopal complex is *shellable* if it is pure and has a shelling.

Thus shellability is not defined for complexes that are not pure, although it turned out in recent work by Björner & Wachs [99] that an extension of the concept to nonpure complexes is possible and useful.

Examples 8.2.

- (i) Every 0-dimensional complex is shellable, by definition. A 1-dimensional complex (a graph) is shellable if and only if it is connected. In particular, this means that it is pure (i.e., has no isolated vertices). A shelling order is an ordering of the edges $e_1, e_2, \dots, e_s$ in such a way that the set $\{e_1, \dots, e_j\}$ describes a connected subgraph for every j : this comes from the condition that the intersection of the edge e_j with the earlier edges has to be 0-dimensional, and thus nonempty.
- (ii) The following are three 2-complexes in the plane $\mathbb{R}^2$.



The first two are not shellable, but the third one is. (Check this!) In each of them, there is a beginning of a shelling indicated, that is, the complex given by the numbered facets together with their faces is shellable. However, if you try to add the last facet in any of the first two examples, you violate condition (ii) of Definition 8.1.

- (iii) Every simplex is shellable, and every ordering of its facets is a shelling order. This immediately follows by induction on the dimension, since the intersection of F_j with F_i ($i < j$) is always a facet of F_j in this case.
- (iv) The d -cubes are shellable: by induction on dimension one can show that every ordering of the $2d$ facets $F_1, F_2, \dots, F_{2d}$ such that the first and the last facet are opposite, $F_1 = -F_{2d}$, is a shelling order. (The condition $F_1 = -F_{2d}$ is sufficient, but not necessary, see Exercise 8.1(i)!)
- (v) The pile of cubes $\mathcal{P}_d(a_1, \dots, a_d)$, see Example 5.4, is shellable for arbitrary finite $a_i \geq 1$. For this, we use part (iv) to see that the lexicographic order on the little cubes in the pile is a shelling order.

Remarks 8.3.

- (i) We will see in the next section that condition 8.1(i) is in fact redundant: the boundary complex $\mathcal{C}(\partial F_1)$ of *every* polytope is shellable. However, if one defines shellability more generally for cell complexes rather than polytopes, as in Björner [85], then this is necessary.
- (ii) For *simplicial* complexes condition 8.1(i) is redundant because of Example 8.2(iii). In this situation, condition 8.1(ii) can also be simplified considerably: it can be replaced by

8.1(ii') For $1 < j \leq s$ the intersection of the facet F_j with the previous facets is nonempty and pure $(k-1)$ -dimensional.

In other words, for every $i < j$ there exists some $l < j$ such that the intersection $F_i \cap F_j$ is contained in $F_l \cap F_j$, and such that $F_l \cap F_j$ is a facet of F_j .

(iii) One might be tempted to weaken condition 8.1(ii), and only require the following:

8.1(ii'') For $1 < j \leq s$ the intersection of the facet F_j with the previous facets

$$F_j \cap \left(\bigcup_{i=1}^{j-1} F_i \right) = G_1 \cup \dots \cup G_r$$

is nonempty, pure $(k-1)$ -dimensional, and shellable.

This can be done. It yields the original definition by Bruggesser & Mani [139], which is weaker than Definition 8.2. Although the main conclusions of shellability one wants remain valid, this weaker version does not have the nice combinatorial characterization that is possible for the stronger version [98].

We continue, with more examples.

Examples 8.4.

- (i) Every polytopal subdivision of a 2-polytope is shellable — see Exercise 8.0.
- (ii) The boundary of every 3-polytope is shellable. This follows from (i): for this we first shell a Schlegel diagram $\mathcal{D}(P, F)$ of P , which is a subdivision of a 2-polytope. This corresponds to a shelling of the whole boundary ∂P except for the facet F . The shelling can be completed by taking F as its last facet.

Subdivisions of 2-polytopes and boundaries of 3-polytopes are easy to shell. One reason is that no matter how we start the shelling, we can't get stuck. Let's introduce some fancy terminology for this.

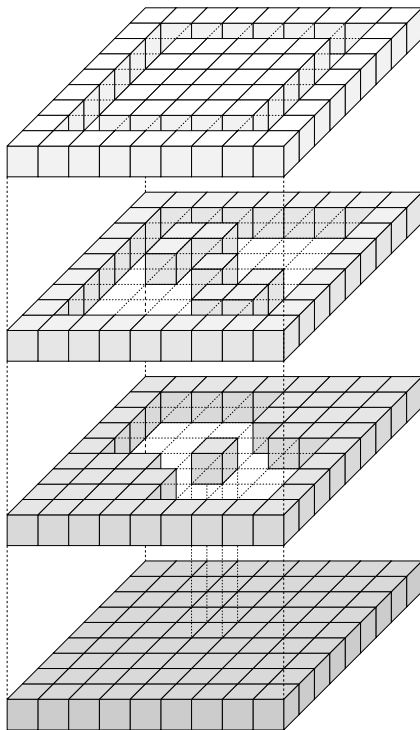
Definition 8.5. (Danaraj & Klee [172, p. 37])

A polyhedral complex is *extendably shellable* if every partial shelling can be continued, that is, if for every shellable subcomplex of the same dimension there is a shelling of the whole complex that shells the subcomplex first.

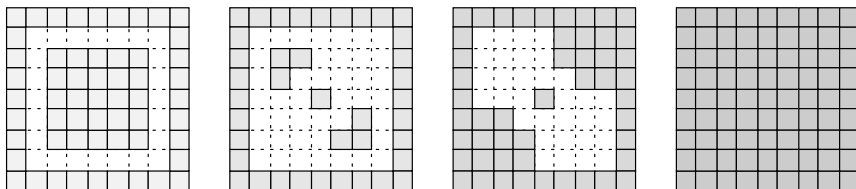
In this sense, we see from Examples 8.4 that subdivisions of 2-polytopes as well as the boundaries of 3-polytopes are extendably shellable. Similarly, one can show that the d -cubes are extendably shellable (Exercise 8.1(i)).

Lemma 8.6. *The pile of cubes $\mathcal{P}_3(9, 9, 4)$ is not extendably shellable.*

Proof. For this, consider the picture below. It depicts a subcomplex of the pile, broken into layers (so you can see what happens inside). The subcomplex is easily seen to be shellable (for example, first shell the bottom layer, then the next layer and the walls, then add the central axis, then the top layer of the interior part, then the six remaining cubes).

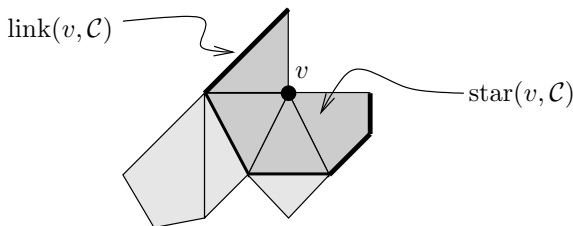


A different way to draw this type of complex (more abstractly) shows the four layers, top to bottom, from left to right.



If we try to add any new cube of the pile $\mathcal{P}_3(9, 9, 4)$ to the subcomplex, then the intersection with what is already there is not pure 2-dimensional, or not connected, or both. Thus, this is a shellable part of $\mathcal{P}_3(9, 9, 4)$, and we are stuck: so $\mathcal{P}_3(9, 9, 4)$ is not extendably shellable. $\square$

For a polyhedral complex $\mathcal{C}$, define the *star* $\text{star}(v, \mathcal{C})$ of the vertex v to be the polytopal subcomplex of all faces that contain v , and their faces. Let the *link* be the subcomplex $\text{link}(v, \mathcal{C})$ of all faces $G \in \text{star}(v, \mathcal{C})$ of the star that do not have v as a vertex.



If $\mathcal{C}$ is pure of dimension d , then so is $\text{star}(v, \mathcal{C})$, and $\text{link}(v, \mathcal{C})$ is then pure of dimension $d-1$.

The following lemma is a quite trivial, but important, piece of information about the “local structure” of shellable simplicial complexes. For nonsimplicial complexes, it becomes nontrivial (if it is true: Problem 8.4*).

Lemma 8.7. *Let $\mathcal{C}$ be a shellable simplicial complex, with shelling order $F_1, F_2, \dots, F_s$. Then the restriction of this order to $\text{star}(v, \mathcal{C})$ yields a shelling order for the star, and also for $\text{link}(v, \mathcal{C})$.*

Proof. We directly verify condition 8.1(ii'): let F_j be a facet in the star (so $v \in F_j$), and let F_i be an earlier facet that also lies in the star (with $i < j$ and $v \in F_i$). Since we have a shelling of $\mathcal{C}$, there is a facet F_l with $l < j$ such that $F_i \cap F_j \subseteq F_l \cap F_j$. But this implies that $v \in F_l$, so F_l is in the star of v .

The same proof also shows that we get a shelling of the link, since (in the simplicial case) we have a bijection between k -faces $G \in \text{link}(v, \mathcal{C})$ and $(k+1)$ -faces $\widehat{G} = \text{conv}(G \cup v) \in \text{star}(v, \mathcal{C})$. $\square$

Theorem 8.8. (Rudin [468])

The 3-simplex Δ_3 can be triangulated in a nonshellable way.

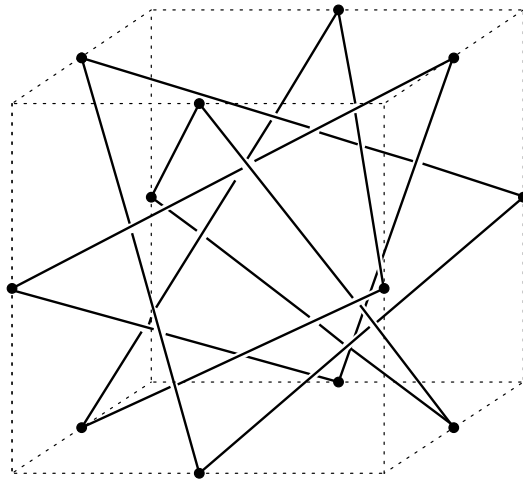
The first nonshellable triangulation of a tetrahedron (with 14 vertices, 41 facets, all vertices on the boundary) was constructed by Rudin [468] in 1958. Her construction is subtle and hard to visualize, and it seems to be the only one in the literature. So, instead of reproducing it, here is a different construction that shows that the tetrahedron and the 3-cube have nonshellable triangulations.

Example 8.9 (The Danzer cube). Let C_3 be our standard cube in $\mathbb{R}^3$, and let V^{12} be the set of 12 midpoints of the edges of C_3 , that is, the set of all points in $\mathbb{R}^3$ with one coordinate 0 and the others equal to ± 1 . The twelve points lie on the boundary of the cube C_3 , but also on the boundary

of the tetrahedron

$$\begin{aligned}
 T_3 &:= \left\{ \mathbf{x} \in \mathbb{R}^3 : \begin{array}{l} +x + y + z \leq 2 \\ +x - y - z \leq 2 \\ -x + y - z \leq 2 \\ -x - y + z \leq 2 \end{array} \right\} \\
 &= \operatorname{conv} \left\{ \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ +2 \\ +2 \end{pmatrix}, \begin{pmatrix} +2 \\ -2 \\ +2 \end{pmatrix}, \begin{pmatrix} +2 \\ +2 \\ -2 \end{pmatrix} \right\}.
 \end{aligned}$$

Next we construct a set E^{12} of 12 edges between the points in V^{12} . For this, take one edge that connects the midpoints of two skew edges of the cube, as well as all the images of that edge under the 12 symmetries of the cube which correspond to orientation-preserving symmetries of the tetrahedron T_3 .

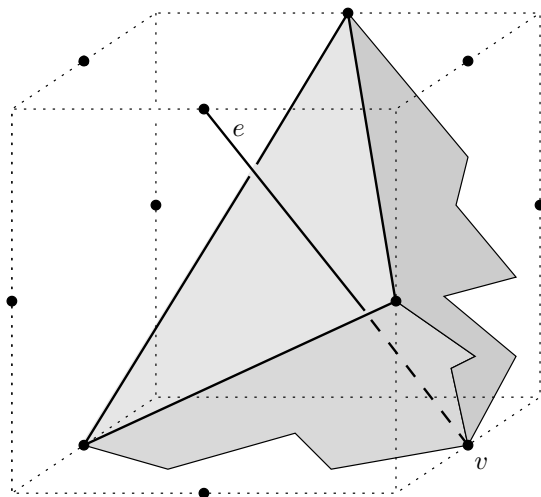


It turns out that the 12 edges we get (see the drawing) are grouped into four disjoint triangles. The key property is that every edge is surrounded by a triangle. By symmetry, it is in fact sufficient to verify this for one single edge $e \in E^{12}$.

The next step is to construct a triangulation of the cube C_3 , respectively of the tetrahedron T_3 , which contains the edges in E^{12} as faces (not subdivided!). While this can be done explicitly, here we resort to a powerful tool: Whitehead's *completion lemma* [560, Thm. 5], according to which every partial triangulation of $\mathbb{R}^3$ can be completed. (In fact, the same is true in $\mathbb{R}^d$, according to Bing [80, Lemma 6]. See Bing [82, Sect. I.2] for a textbook version.) Hence, we can take any triangulation of the boundary of C_3 , respectively T_3 , that uses the vertices in V^{12} , plus the twelve edges in E^{12} , and complete this simplicial complex to a triangulation of C_3 , respectively T_3 . There will be additional interior vertices necessary for this.

Now assume that the resulting simplicial complex $\mathcal{C}$ is shellable. We start with one tetrahedron F_1 , and then add new ones, $F_2, F_3, \dots$. At every step, except possibly the first, not more than one new edge from the set E^{12} can be added, since the edges in one of the triangles cannot be in a common simplex, and if two skew edges were new, then this would contradict a characteristic property of shellings of simplicial complexes: there has to be a unique minimal new face at each step, which is nonempty after the first step (see Exercise 8.2).

Now let F_{j+1} be the tetrahedron by which the *last* edge e from E^{12} is added. At this point of the shelling the complex $\mathcal{C}_j := \mathcal{C}(F_1, \dots, F_j)$ already contains the three edges e_1, e_2, e_3 that surround e . In fact, the circle formed in $\mathcal{C}_j$ by $C := e_1 \cup e_2 \cup e_3$ can be contracted within the complex $\mathcal{C}_j$: this is a property of shellable complexes of dimension at least 2, which is easy to verify by induction (they are “simply connected”).



Since the circle C surrounds the edge e , we have to “pass over” a vertex $v \subseteq e$ when we contract C in $\mathcal{C}_j$. Also, the link of this vertex v in the complex $\mathcal{C}_j$, $\text{link}(v, \mathcal{C}_j)$, is a 2-dimensional shellable complex, by Lemma 8.7. However, we can contract our circle C within $|\mathcal{C}_j|$ until it lies in $\text{link}(v, \mathcal{C}_j)$, but it cannot be contracted within the link, because then it would not pass over v . This shows that the link is not shellable; contradiction. $\square$

8.2 Shelling Polytopes

The complex of a polytope $\mathcal{C}(P)$ is shellable if and only if the boundary complex $\mathcal{C}(\partial P)$ is shellable. Thus, “shelling a polytope” means finding a shelling order for the facets of P . How do we do this? Is there an obvious